Friction adjusts before sliding begins

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Static friction / ES: fricción estática / PT: atrito estático

Before sliding. Adjusts to match the pull, up to a limit.

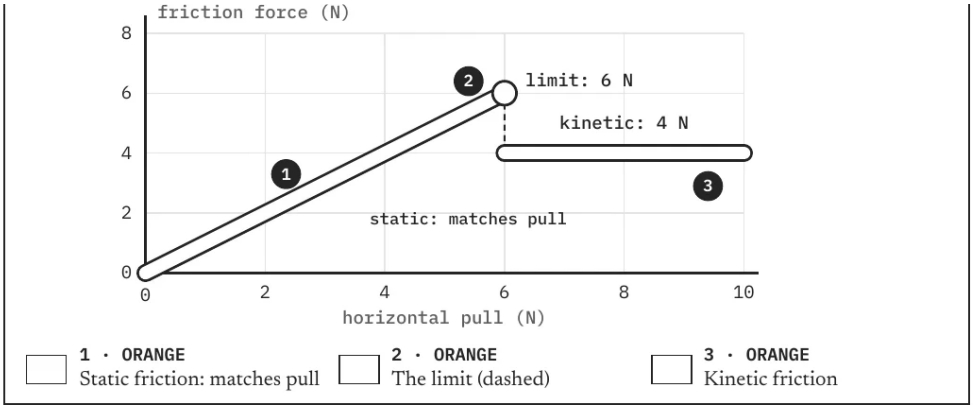
Also on your note page: Maximum static friction, Kinetic friction, Coefficient (μ).

B READ THE RELATIONSHIP

Static friction acts between surfaces that do not slide past each other.

Static friction adjusts up to a limit; sliding friction acts during slip

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 2 kg crate slides right. Pull = 10 N right; kinetic friction = 4 N left. Find its acceleration.

Given m = 2 kg pull 10 N right f k 4 N left

Rule a = F net ÷ m

Work F net = 10 − 4 = 6 N right a = 6 ÷ 2

Answer 3 m/s² right

E SAVE FOR THE FINAL MEETING

Friction needs surfaces rubbing together. In space there are none to slow XJ down, one more reason it keeps its speed between observations.

Read the physics: Static friction

SUPPORTED READING / read one part, then point to the evidence Name

01 PICTURE IT

Push gently on a crate. If it stays still, static friction can balance the pull. A harder push needs more static friction, up to a maximum.

02 HOW IT WORKS

After slipping begins, use kinetic friction. In this model its size is the kinetic coefficient times the normal force. It opposes relative slipping at the contact.

03 WHAT TO CHECK

On a force trace, find the peak just before slip. Compare it with the steady-speed pull while the surfaces slide. Do not use the maximum static force for every pull.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg crate slides right. Pull = 10 N right; kinetic friction = 4 N left. Find its acceleration. 3 m/s² right.

05 CONNECT TO XJ

Friction needs surfaces rubbing together. In space there are none to slow XJ down, one more reason it keeps its speed between observations.

SAY IT BEFORE YOU WRITE

When do you switch from static to kinetic friction? Before sliding, friction ___. During sliding, we use ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Static friction

ES: fricción estática / PT: atrito estático

Meaning in my words	How can you tell?
How would you explain static friction to a friend? In my words, this means ___.	You push a crate gently and it stays still. What force balances your sideways push? I can tell because ___.
Example or sketch	**Non-example and why**
A stationary crate on a level floor is pushed right with 3 N below the sliding threshold. Draw friction. My example shows ___.	The crate is already sliding across the floor. Would the friction at that contact still be static? I would change ___ because ___.

Maximum static friction

ES: fricción estática máxima / PT: atrito estático máximo

Meaning in my words	How can you tell?
How would you explain maximum static friction to a friend? In my words, this means ___.	The floor can supply up to 6 N. How much friction balances a gentle 2 N horizontal push? I can tell because ___.
Example or sketch	**Non-example and why**
With a 6 N limit and no other horizontal forces, what do you predict for a 7 N push? My example shows ___.	A friend writes 6 N for every trial, even with no sideways push. What does 6 N mean? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Kinetic friction

ES: fricción cinética / PT: atrito cinético

Meaning in my words	How can you tell?
How would you explain kinetic friction to a friend? In my words, this means ___.	A crate slides right across a stationary floor. Which way does kinetic friction act on the crate? I can tell because ___.
Example or sketch	**Non-example and why**
If the coefficient is 0.2 and the normal force is 10 N, what arrow would you draw? My example shows ___.	A crate stays put while you push gently. Should you use the kinetic-friction rule? I would change ___ because ___.

Coefficient (μ)

ES: coeficiente de fricción / PT: coeficiente de atrito

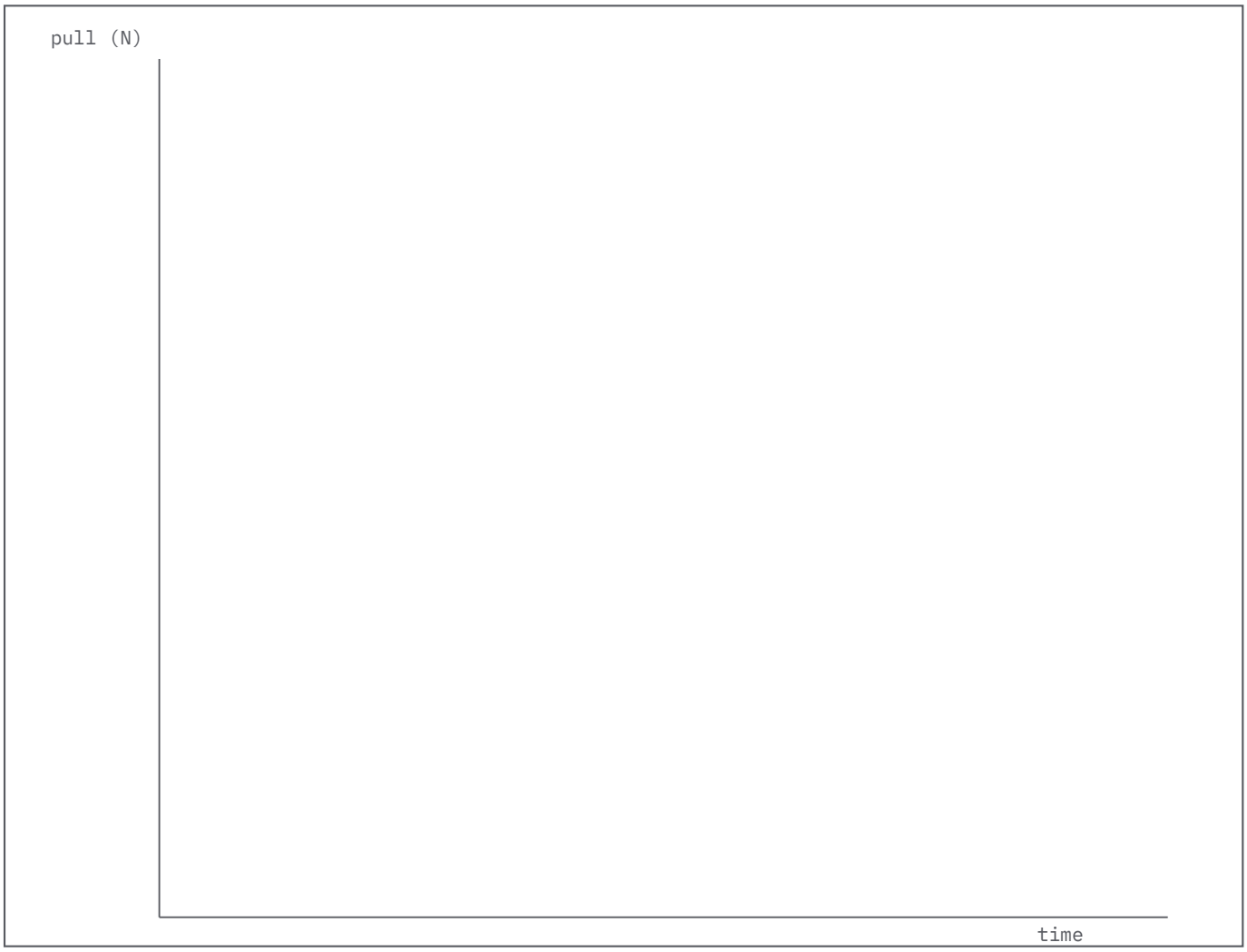
Meaning in my words	How can you tell?
What does the coefficient of friction tell us about two surfaces? In my words, this means ___.	A sliding block has 2 N friction and 10 N normal force. How do you find the kinetic coefficient? I can tell because ___.
Example or sketch	**Non-example and why**
With the same surfaces and a 20 N normal force, what sliding friction would μ = 0.2 predict? My example shows ___.	A student writes “μ = 0.2 N.” What happens to the units in the ratio? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a force trace. Label the peak just before slip and the steady-speed pull while the surfaces slide.

Labels you can use: at rest · just before slip · sliding.



Tell a partner

Before sliding, friction ___. During sliding, we use ___.

Cut from the table to space

17

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use contact surfaces to decide when friction acts.

CHOOSE ONE JOB TOGETHER

EFFECTS ENGINEER ☐

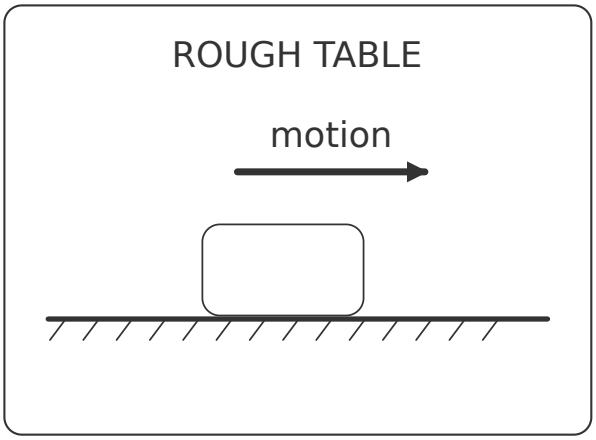
Set friction for each scene. Mark the touching surfaces and explain what could change XJ's motion in empty space.

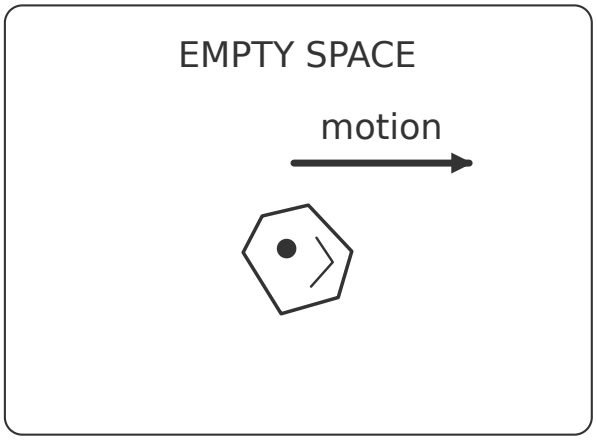
SCIENCE BROADCASTER ☐

Correct the caption for the two shots. Point to the surfaces and say what could still bend XJ's path.

Word help: contact / friction / drag

A show cuts from a block sliding on a rough table to XJ coasting in empty space. The proposed caption says “Both will stop from surface friction.”





01 OUR FIRST CALL

What should the two-scene caption say? Explain using the contact surfaces.

In this scene, ___ touches ___, so ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What changed in the new shot? Update the force explanation for XJ.

Now we would say ___ because ___.

Switch speakers. Which surfaces are touching?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks